

The Future Unemployed

The NEXT WAVE IN TECHNOlogy

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Figure 1: Team Logo

Project Overview

Description

A client from a group of wealthy investors has approached us to create a system that simulates Panther Cloud Airlines' fleet operations over a two-week period. This platform will track and display real-time flight paths for each aircraft, showing their current positions and routes as they operate throughout the day. The system will calculate profitability for individual flights and routes by factoring in ticket sales, fuel costs, airport fees, and maintenance expenses. It includes a passenger load forecasting feature that predicts how full each flight will be based on the population of the metropolitan area. Additionally, the platform has an integrated booking system where users can search for flights, view available seats, and make reservations. All these components work together in real-time, when someone books a flight, the passenger count updates automatically, which then affects the profitability calculations, giving investors a complete view of the airline's operational performance and financial health.

Team Positions

- Reid Holloway - Scrum Master/Project Leader
- Graham McCabe - Configuration/Quality Assurance Manager
- Gracie Mitchell - Documentation Specialist
- Preston Frost - Principal Developer/Trainer

Work Breakdown

Reid Holloway

- Calculated data for flight physics for take off, landing, and cruise
- Calculated distances for each airport
- Created 14 week schedule for the project

Graham McCabe

- Created rough draft UI design and defined user types
- Researched and determined profitability tracking and calculations
- Identified project risks and constraints

Gracie Mitchell

- Created Project Plan Presentation
- Maintain documentation and organization in the google drive
- Collected data for airport distances

Preston Frost

- Researched Hardware and Software Requirements
- Created database using MariaDB
- Researched how to plan for future backend implementation

Project Goals

Financial Analysis

- Identify which routes and aircraft are most profitable to guide investment decisions
- Determine optimal pricing strategies across different fare classes and routes
- Project overall revenue and profit margins for the two-week operational period

Operational Efficiency Goals

- Optimize aircraft utilization rates to maximize flight hours and minimize ground time
- Identify underperforming routes that may need schedule adjustments or discontinuation
- Analyze fuel efficiency across different aircraft types and routes

System Demonstration Goals

- Showcase the technical capabilities of real-time simulation and integrated booking systems

Project Scope

- Part 1
    - Create a comprehensive timetable for an airline fleet operating across:
        - Top 30 US airports (FAA authorized)
        - 1 international route: New York ↔ Paris (daily)
    - Fleet:
        - 15 Boeing 737-600s
        - 15 Boeing 737-800s
        - 12 Airbus A220-100s
        - 13 Airbus A220-300s
- Part 2
    - Simulating 14-days of operation with different variables
        - Weather delays, icing delays, mechanical failures
        - Flight cancellations, gate delays
        - Jet stream effects
        - Track all passenger movements, aircraft positions, gate usage
    - Capabilities
        - Flight scheduling with realistic constraints
            - Gates
            - Maintenance
        - Route optimization based on
            - Distance
            - Range
            - Hub
            - Demand
        - Passenger journey
            - Connections
            - Profit
        - Aircraft data and usual performance
        - Finances
            - Costs
            - Revenue
            - Profit
        - Conversions to and from Paris
    - **NOT INCLUDED**

        - Crew Scheduling
        - Baggage Tracking
        - Multi-year simulation
        - Interaction with other airlines
        - Regulated airspace
    - Criteria for success
        - All 55 aircraft scheduled properly
            - Not forgetting gate numbers or passengers
        - All airports have ways from one to another
        - All days simulate with no issues
        - Profit is accurately output

Project Design

Hardware and Software Requirements

Hardware

- Model: Dell Pro 14 Premium
- Processor: Intel Core Ultra 7 268V, 2200 MHz, 8-core processor
- Memory: 32 GB of RAM
- Storage: 120 GB of available disk space
- Display: 14-inch monitor

Software

- Operating System: Microsoft Windows 11 Enterprise
- Execution Environment: Website using HTML and REACT Database Management: MariaDB, deployed via Docker
- Version Control: GitHub
- Development Tools: Makefiles, Trello/Monday.com for project management, and standard linux compilers

Databases

System Architecture & Environment

- DBMS: MariaDB will be used as the primary database engine
- Deployment: Uses Docker for portability and Docker Compose for ease of deployment
- Persistence: Persistent volumes will maintain data between container executions

Data Entities

- Static Entities: Fleet data, Airport data, and complete timetable

- Dynamic Data (simulation): Gate assignments, flight paths, and scheduled v.s. actual time
- Financial Records: Profitability overview, fuel costs, and landing fees

Permissions (Access Control)

- Gate Agent (Employee): Write booking records and view schedule
- Airport Administrator: Read financial reports and profitability tables.

UI

User Roles

Role	Permissions
Airport Employee	Log in, view scheduled flights, book flights for customers
Airport Administrator	All employee permissions plus access to route and plane profitability reports

Screen Descriptions

- Login Screen
  - Simple authentication form with email and password fields, plus error handling for invalid credentials

- Airplane Schedule
  - Displays a flight schedule table with a "Book Flight" and “Administrator Dashboard" button.
    - The “Book Flight” action that opens a booking form for entering customer details.
    - The “Administrator Dashboard” button is only accessible to Administrators and takes the user to the Administrator Dashboard

- Administrator Dashboard
  - Provides access to profitability analytics with two reporting views:
    - Route Profit Table Displays profit or deficit for each route, filtered by start and end locations

        - Plane Profit Table – Shows Plane #, total number of flights, and profit or deficit per aircraft

- Book Flight
    - Allows users to search for available flights by entering Date of Departure, Date of Return, Departure location, and Arrival destination. Results are displayed in a table with Destination, Flight #, Departure Time, and Status, with a "Book Flight" button to proceed.

- Confirm Route
    - Presents the selected flight details for final review, including Departure, Arrival, Board Time, Arrival Time, and Flight #. If multiple flights match the search criteria, each option is displayed for the user to select before confirming the booking.

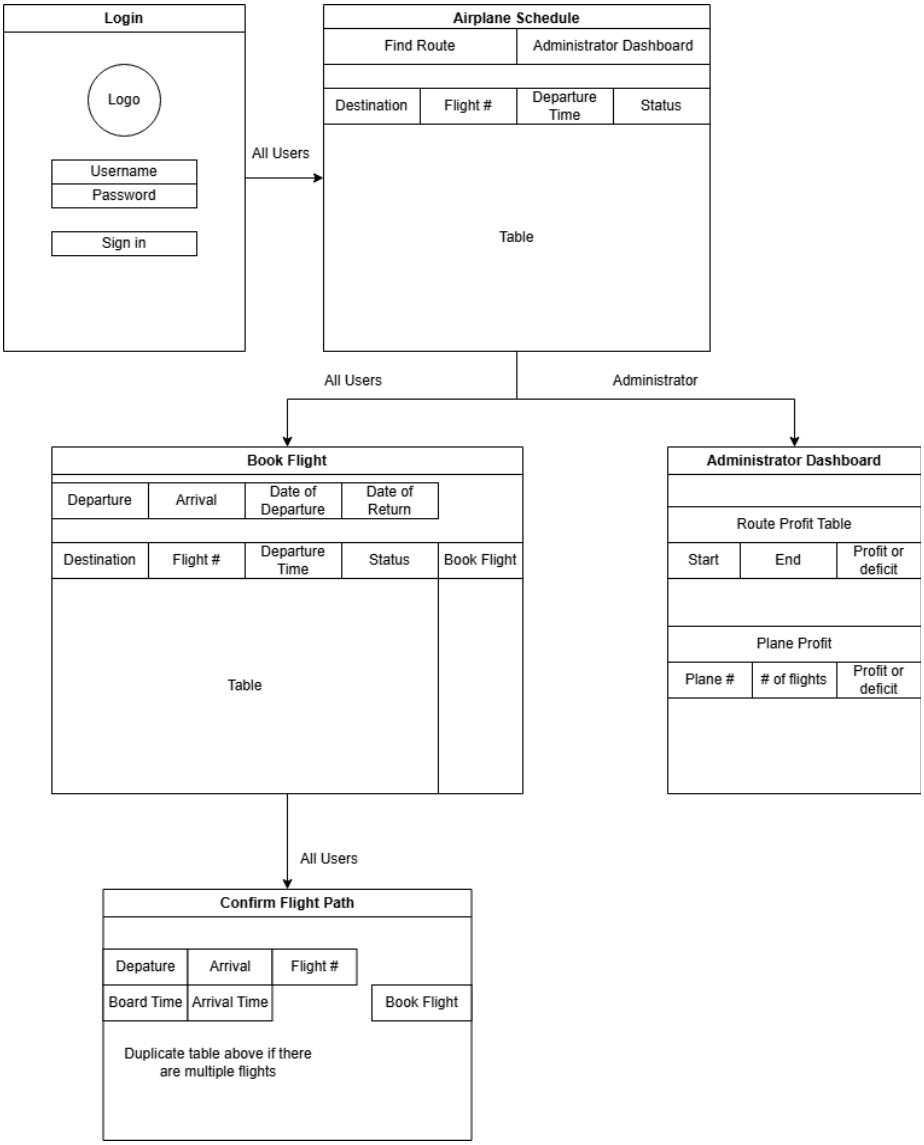


Figure 2: UI Diagram

Project Schedule

Timeline

- Our timeline will be put into 5 phases

Phase	Duration	Goal
Phase 1: Setup and Planning	Week 1-2	Establish a foundation and gather necessary data
Phase 2: Flight Scheduling	Week 3-7	Create flight timetable and database
Phase 3: Simulation	Week 8-11	Build and test a working simulation
Phase 4: Testing	Week 12-13	Confirm documentation in system
Phase 5: Presentation/Delivery	Week 14	Final presentation and submission

Table 1: Timeline Phases

Key Milestones

Week	Milestone	Criteria
2	Complete Data Collection	All data collected for all 30 airports and Paris, hubs decided
3	Database is Operational	Docker is up and running, schema created
4	Flight Physics Working	Flight times are calculated between any

		two airports
7	Part 1 Done	Complete timetable done and searchable
8	Simulation and UI Framework	Can simulate a full day with no delays or severe flaws
11	Part 2 Done	14-day simulation successful for most all scenarios
13	Documentation Complete	System installable on correct machine and README file complete
14	Final Delivery	Presentation and Deliverables Submitted

Table 2: Key Milestones

Dependencies

Key Prerequisites per Phase

- **Week 3**
  - Airport data collected
  - Coding language selected
  - GitHub and Database created
- **Week 7**
  - Database fully up and working
  - Flight physics complete
  - Flight scheduling algorithm complete
- **Week 11**
  - Part 1 Timetable complete
  - Simulation framework and UI working correctly
  - All scenarios tested
- **Week 14**
  - All testing complete
  - Documentation finalized
  - Practice presentation and confirm project files

Deliverables

Completed Deliverables

Phase 1

- **Week 1**
    - Private GitHub repository
    - Physical journals (one per team member)
    - Directory structure
    - Project board
    - Language selection documentation
- **Week 2**
    - Complete airport database
        - All 30 + Paris
            - Distances
            - Metro populations
            - Hubs
            - Gate calculations
            - Time from airport to airport
- Project Records
    - Journals kept up with per meeting
    - Design decisions
    - Meeting Minutes and notes

Upcoming Deliverables

Phase 2

- Database
    - MariaDB database fully filled out
    - Complete database schema
- Scheduling Engine
    - Wind speed effect at cruising altitude
    - Gate management system
    - Aircraft assignment
    - Route algorithm
    - Maintenance schedule
    - Passenger demand model
- User
    - Complete flight timetable
    - Passenger flight profit
    - Printed time table

- Documentation
    - Schedule of meetings
    - Algorithm design
    - Cost and revenue calculations
- Project Records
    - Journals kept up with per meeting
    - Design decisions
    - Meeting Minutes and notes

Phase 3

- Simulation Framework
    - Passenger database tracking
    - Confirm every passenger that flies with us gets to destination
    - Gate usage tracker
    - Aircraft distance to maintenance
- Disruption Scenarios
    - Weather
    - Ice
    - Gate numbers
    - Wind
- Query and Database
    - Financial performance collected
        - # of passengers transported
        - Operating costs
        - Revenue
        - Profit
- Simulation
    - 14 day simulation database working
    - All flights work
    - Most financial routes
- Project Records
    - Journals kept up with per meeting
    - Design decisions
    - Meeting Minutes and notes

Phase 4

- Testing Deliverables
    - Cross-platform testing
    - System validation
    - Bug fixes
- System Documentation
    - README file

    - Database set up
    - APIs
    - Makefile
- User Documentation
    - User manual
    - Flight search interface
    - Query interface
    - Troubleshooting
- Project Records
    - Journals kept up with per meeting
    - Design decisions
    - Meeting Minutes and notes

Phase 5

- Presentation Materials
    - 20-minute presentation covering
        - Project overview + objectives
        - System architecture
        - Flight scheduling demonstration
        - Simulation demonstration
        - Results and financial analysis
        - Conclusions
    - Live system demonstration
    - Backup demonstration materials
        - Screenshots
        - Screen recordings
- Final Submission
    - Complete source code on GitHub
    - All documentation (user and technical)
    - Installation package + makefile
    - Simulation results database
    - Financial report and explanation
    - Completed journals (one per team member)
    - Final report
    - All meeting minutes

Project Constraints and Risks

Constraints

ID	Constraint	Impact
C1	Scheduling	Coordination conflicts may delay milestone completion and team meetings
C2	Time	Limited development window restricts feature scope and testing thoroughness
C3	Memory	System resource limitations may restrict simulation complexity and concurrent operations
C4	Available Tools	Restricted access to certain software/API may require workarounds or feature modifications
C5	Performance	Processing Constraints may limit real time calculations and user capacity

Table 3: Constraints

Risks

ID	Risk	Impact
R1	Weather	Severe conditions could prevent team members from meeting or accessing campus resources
R2	Power or Wifi Outage	Loss of connectivity disrupts development, testing, and cloud-based collaboration
R3	Removing or Deleting Docs	Accidental file loss could set back progress and require rework

R4	Absence	Team member unavailability delays assigned tasks and increases workload on others
R5	Ability of Delivering Individual Job	Skill gaps or underperformance affects dependent tasks and overall timeline
R6	Miscommunication	Misunderstandings lead to duplicated effort, incompatible components, or missed requirements
R7	Falling Behind on Documentation	Incomplete records complicate future maintenance and grading evaluation
R8	System Crashing	Unexpected failures cause data loss and interrupt development workflow
R9	Accessibility	Platform incompatibilities or limited access may exclude users or team members

Table 4: Risks